The circumference of a circle is $2\pi r$ *units*, which can also be written as πd *units*. As $\pi = 3.14$ (approximately) and it is slightly more than 3, for some quick guess, we shall say that a circle whose diameter is *d units* shall have its circumference slightly more than three times its diameter.



For example,

If a round table of diameter of 3*m* is to be decorated by flower strings around it, it will require a little more than 9*m* of flower strings.

2.2 Parts of a Circle

A circle is the path traced by a moving point so that its distance from a fixed point is always constant. The fixed point of the circle is called its **'centre'** and the constant distance is called its **'radius'.**

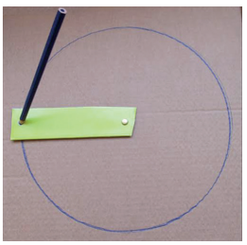


Fig. 2.1

Further, if any two points on a circle are joined by a line segment, then the line segment is called a **'chord'.** A chord divides a circle into two parts. A chord which passes through the centre of a circle is called as a **'diameter'.** A diameter of a circle divides it into two equal parts. It is also the 'longest chord' of a circle.

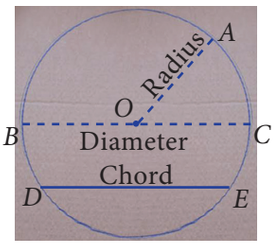


Fig. 2.2

Activity

1. Using a bangle, draw a circle on the paper and cut it. Then mark any two points A and B on it and fold the circle so that the fold has A and B on it. Now, this line segment represents a chord.
2. By paper folding, find two diameters and hence the centre of a circle.
3. Check whether the diameter of a circle is twice its radius.

2.2.1 Circular Arc and Circular Sector

Look at the glass bangle and the pizza given in Fig. 2.3 carefully.



Fig. 2.3

Though both are of circular shapes, what we understand is that, the bangle indicates the boundary of the circle, whereas the pizza indicates the plane enclosed within the boundary of the circle. Thus, it is clear that the bangle indicates the circumference of the circle whereas the pizza indicates the area of the circle.

From them, cut a part as shown in Fig. 2.4.

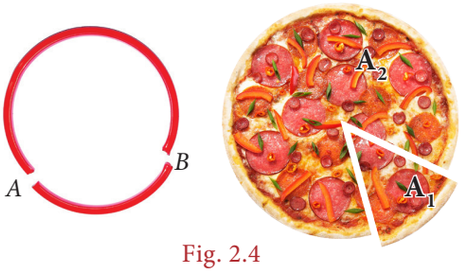


Fig. 2.4

Each of the glass bangle pieces represents the circular arcs. Here, arc AB $\left(\overset{\frown}{AB}\right)$ is the smaller one and

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